Solution:

Year x	No. of arrivals y_i	$X = x_i - 2005$	$u_i = 2X - 5$	u_i^2	$u_i y_i$
2005	12	0	–5	25	–60
2006	13	1	–3	9	–39
2007	18	2	–1	1	–18
2008	20	3	1	1	20
2009	24	4	3	9	72
2010	28	5	5	25	140
	$\sum_{i=1}^{6} y_i = 115$		$\sum_{i=1}^{6} u_i = 0$	$\sum_{i=1}^{6} u_i^2 = 70$	$\sum_{i=1}^{6} u_i y_i = 115$

The equation of the straight line is $y = a + bx$

$= a + bu$ where $u = 2X - 5$

Using the normal equation we have,

$$a = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{115}{6} = 19.17$$

$$b = \frac{\sum_{i=1}^{n} u_i y_i}{\sum_{i=1}^{n} u_i^2} = \frac{115}{70} = 1.64$$

$y = a + bu$

$= 19.17 + 1.64\ (2X - 5)$

$= 19.17 + 3.28X - 8.2$

$= 3.28X + 10.97$

That is, $y = 10.97 + 3.28X$

To get the required trend values, put $X = 0, 1, 2, 3, 4, 5$ in the estimated equation. Thus,

$X = 0, y = 10.97 + 0 = 10.97$

$X = 1, y = 10.97 + 3.28 = 14.25$

$X = 2, y = 10.97 + 6.56 = 17.53$

$X = 3, y = 10.97 + 9.84 = 20.81$

$X = 4, y = 10.97 + 13.12 = 24.09$

$X = 5, y = 10.97 + 16.4 = 27.37$

Hence, the trend values for 2005, 2006, 2007, 2008, 2009 and 2010 are 10.97, 14.25, 17.53, 20.81, 24.09 and 27.37 respectively.

(ii) Seasonal variation

Seasonal variations are fluctuations within a year over different seasons.

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